MEASUREMENT VALIDATION
FOR WASTEWATER TREATMENT

SECOND *opinion*

Spot unusual measurements, compare the evidence, and decide what to check first.

PRODUCT
INVESTIGATION SUPPORT

FOR
WASTEWATER PLANT OPERATORS

COVERS
SIX MEASUREMENT GROUPS

01 // PROBLEM

NO LABELS, MANY READINGS

Which unusual reading is *worth checking?*

NO LABELS

The data does not tell us which readings are real problems.

MORE THAN LIMITS

A reading can be unusual for its own point even when it is within a fixed limit.

REVIEW TAKES TIME

Every unnecessary investigation costs an operator attention and time.

28

days of a point's own history decide what is usual for it

NO COMPARISON BETWEEN PLANTS

02 // END-TO-END PIPELINE

HISTORICAL, DAY-LEVEL DATA

From measurements *to review.*

01 // INPUT

PREPARE

Read the Z-info export. Select six exploratory measurement groups and preserve source and point context.

02 // DETECT

SCORE

Compare each point with its own history; apply simple data-quality rules.

03 // ORGANISE

GROUP

Combine flagged days into cases and rank them within each plant.

04 // REVIEW

INVESTIGATE

Show the evidence, unscored reasons and optional AI scores to support review.

IN PARALLEL

Isolation Forest offers an experimental second ranking; the history detector works on its own.

03 // HISTORY DETECTOR

EACH POINT KEEPS ITS OWN HISTORY

Learn what is *usual here.*

01 // BASELINE

LOOK BACK

Use the previous 28 calendar days, with at least 14 valid days. Find the median and robust spread.

02 // EVIDENCE

CHECK TODAY

Measure deviation from the baseline. Also flag stuck values, unexpected zeros and negative values.

03 // OUTPUT

FORM CASES

Group nearby flagged days. Rank by rule hits, duration and peak score; preserve reasons and source references.

SAFEGUARD

Missing, duplicate or insufficient history can leave a reading unscored; unscored is not normal.

04 // EXPERIMENTAL AI LAYER

NO PEER COMPARISON

Spot an unusual *combination.*

01 // FEATURES

SIX SIGNALS

Deviation, daily change, recent deviation, relative level, zero and repeated days.

02 // MODEL

ONE PER GROUP

One model per measurement group learns unusual combinations of history-based features.

03 // ROLE

SECOND VIEW

Compare its ranking with the history detector. It does not replace the primary case queue.

INTERPRETATION

A higher score means a rarer feature pattern, not a fault probability or diagnosis.

05 // EVALUATION

CHRONOLOGICAL AND HONEST

Compare *without guessing.*

TRAIN

THROUGH 2023

Fit each forest using earlier data and a fixed random seed.

VALIDATE

2024

Set the AI alert threshold on validation scores, then keep it fixed.

HOLD OUT

2025

Compare both rankings on the same eligible readings and review budget.

LEARN

REVIEW

Report coverage and workload; expert feedback is needed to judge usefulness.

LIMITS

No confirmed labels: no real-world accuracy or false-alarm savings claim.

06 // OPERATOR DECISION

EVIDENCE, NOT A DIAGNOSIS

Check what *matters first.*

SEE

Review ranked history-based cases.

UNDERSTAND

Read the baseline, rules and source context.

COMPARE

Consult the optional AI ranking.

DECIDE

An operator investigates and records feedback.

CASE VIEW AND FEEDBACK WORKFLOW ARE PROPOSED; THE OPERATOR DECIDES.